Orthodontic Growth Patterns in Children

Advanced Data Analysis

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Outline

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Introduction

The data used in this study come from the Orthodont dataset in the nlme package in R. The dataset contains measurements of the distance between two points in the mouth that are important for studying jaw growth in children. Each child was measured several times at ages 8, 10, 12, and 14 years. Because the same child was measured repeatedly, the observations are correlated. Therefore, a linear mixed-effects model is used to analyze the data. The main purpose of this study is to understand how this distance changes as children grow older and to determine whether the growth pattern differs between boys and girls.

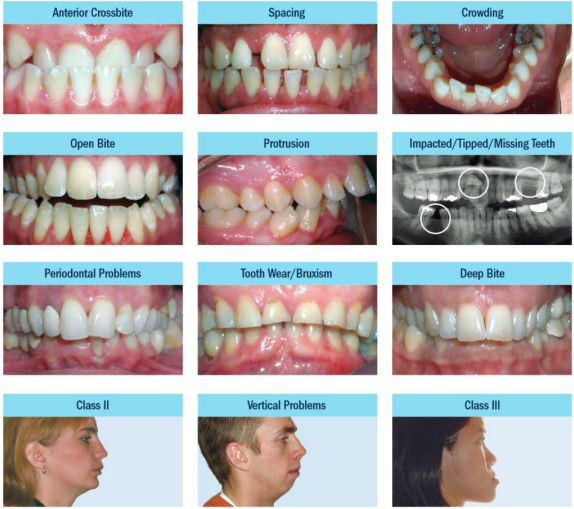

Figure 1: Orthodontic

Baseline Characteristics

Characteristic	Value
Number of children	27
Number of boys	16
Number of girls	11
Baseline age	8 years
Number of repeated measurements per child	4
Observed ages	8, 10, 12, 14 years
Outcome variable	Distance

The baseline measurement was taken at age 8 years for all children.

Exploratory Data Analysis

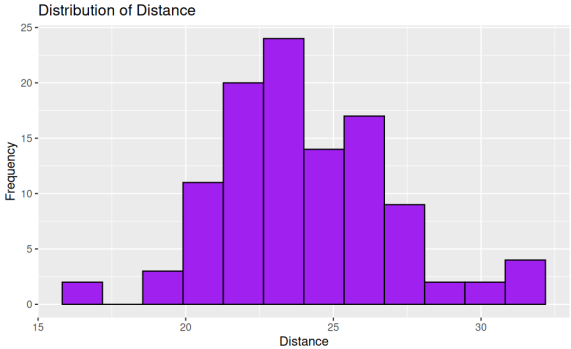

Figure 2: Distribution of Distance

- The distance values are reasonably spread.
- The distribution is approximately symmetric.
- This supports the use of a linear mixed model.

Distance by Sex

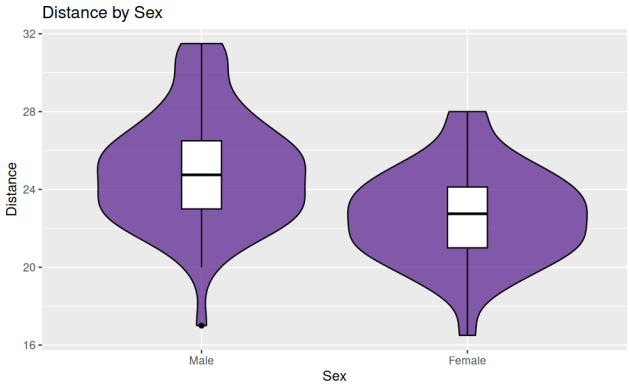

Figure 3: Distance by Sex

Interpretation

- Boys tend to have higher distance values than girls.
- This suggests that sex may be an important predictor.

Distance by Age and Sex

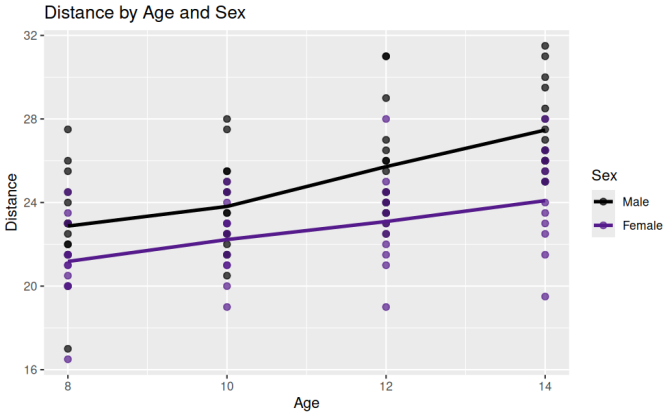

Figure 4: Mean Distance by Age and Sex

Interpretation

- Distance tends to increase with age.
- Boys have higher average distance than girls at most ages.
- Age and sex both appear important.

Model Building

M1 Random intercept model with age and sex

M2 Random intercept model with age, sex, and age×sex interaction

M3 Random intercept and random slope model with age, sex, and interaction

Model M1

$$distance_{ij} = \beta_0 + \beta_1 age_{ij} + \beta_2 Sex_i + u_{0i} + \epsilon_{ij}$$

Model M2

$$distance_{ij} = \beta_0 + \beta_1 age_{ij} + \beta_2 Sex_i + \beta_3 (age_{ij} \times Sex_i) + u_{0i} + \epsilon_{ij}$$

Model M3

$$distance_{ij} = \beta_0 + \beta_1 age_{ij} + \beta_2 Sex_i + \beta_3 (age_{ij} \times Sex_i) + u_{0i} + u_{1i} age_{ij} + \epsilon_{ij}$$

Model Comparison

Table 2: Comparison of Candidate Models

Model	AIC	BIC	logLik
M1: RI + age + Sex	444.856	450.267	-217.428
M2: RI + age * Sex	440.639	456.732	-214.320
M3: RI + RS + age * Sex	443.806	465.263	-213.903

Likelihood Ratio Tests

- M1 vs M2: $p = 0.126$
- M2 vs M3: $p = 0.659$

Model Choice

Although Model 2 had the lowest AIC, Model 3 was selected as the final model because it allows each child to have both a different baseline distance and a different growth rate over age. This is more realistic for child growth data.

Final Model Results

Table 3: Fixed Effects from the Final Model

Term	Estimate	SE	t-value	p-value
Intercept	16.3406	1.0185	16.0433	0.0000
age	0.7844	0.0860	9.1207	0.0000
SexFemale	-1.0321	1.5957	-0.6468	0.5237
age:SexFemale	-0.3048	0.1347	-2.2624	0.0264

Random Effects

- Random intercept variance = 5.786
- Random slope variance = 0.0335
- Residual variance = 1.716

Interpretation of Fixed Effects

Main Findings

- **Intercept:** This is the expected distance for the reference group, that is, **boys at age 0**.
- **Age effect:** Distance increases significantly with age.
- **Sex effect:** Girls have a lower average distance than boys at the reference age, but the main effect is not significant.
- **Interaction effect:** The age$\times$sex interaction is significant, meaning the growth rate differs between boys and girls.
- Therefore, age and the interaction between age and sex are important predictors of orthodontic growth.

Interpretation of Random Effects

- The **random intercept** shows that children start at different baseline distances.
- The **random slope** shows that children also differ in how fast distance changes with age.
- This means growth does not follow exactly the same pattern for every child.
- Including both random effects makes the model more realistic.

Estimated Mean Profiles

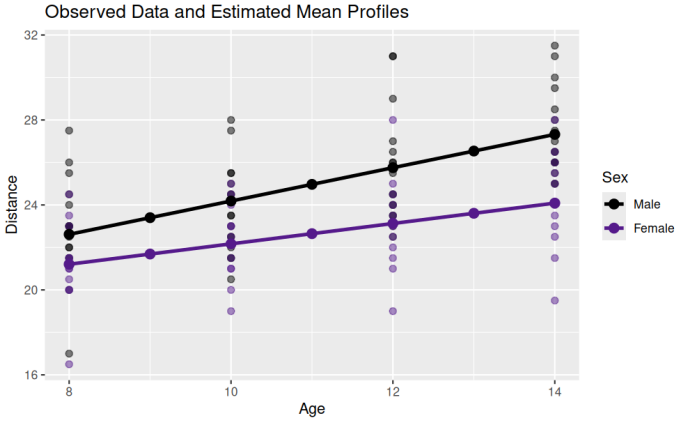

Figure 5: Estimated Mean Profiles by Sex

Interpretation

- Estimated mean distance increases with age for both sexes.
- Boys have higher estimated mean distance across ages.
- Boys show a steeper growth slope over time.

Diagnostic Plots

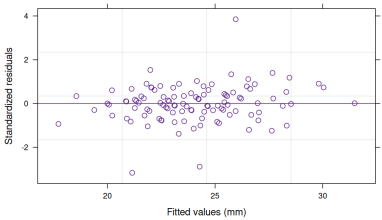

Figure 6: Residuals vs Fitted

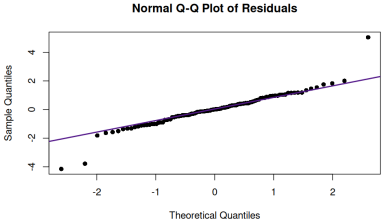

Figure 7: Q-Q Plot of Residuals

Interpretation

- Residuals are scattered around zero with no strong pattern.
- The Q-Q plot is reasonably close to the straight line.
- This suggests that the assumptions of constant variance and residual normality are acceptable.

Random Effects Diagnostics

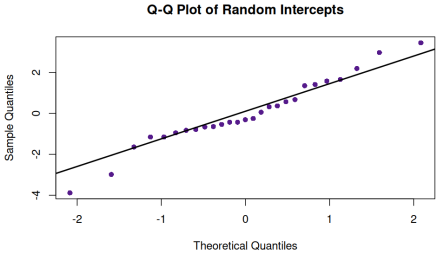

Figure 8: Q-Q Plot of Random Intercepts

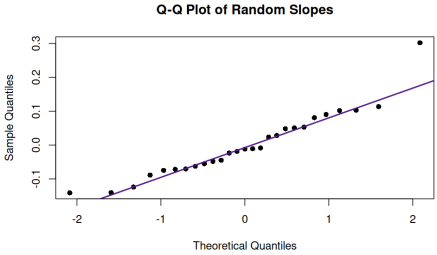

Figure 9: Q-Q Plot of Random Slopes

Interpretation

- The Q-Q plots are reasonably close to the straight line.
- This suggests that the random intercepts and random slopes are approximately normal.
- Therefore, the normality assumption for the random effects is acceptable.

Conclusions

- A linear mixed model was appropriate because the measurements were repeated on the same child.
- Distance increased significantly with age.
- Boys generally had higher average distance than girls.
- The age-by-sex interaction was significant, showing different growth rates by sex.
- The random intercept and random slope model provided a realistic representation of individual growth.
- Diagnostic plots suggested that the final model assumptions were reasonably satisfied.

Thank
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